

Software Version History — ATS Triton Receiver Interface

Current release: v1.27

Last updated: October 2026

The Triton Receiver Interface is the Windows application used to configure ATS SR3xxx acoustic receivers, record detections in real time, download files from the receiver's SD card, filter detection data, and flash receiver firmware.

This document lists what changed in each release. Changes to the detection filters are summarized here and described in full in the companion document, **Detection Filter History**.

A note on version numbers. Releases use a two-digit minor number: v1.16 is "one point sixteen", v1.27 is "one point twenty-seven". There has never been a v1.6 or v1.06.

v1.27 — June–July 2026

A correctness release. Several changes fix real data and communication errors. **We recommend all users update to v1.27.**

Detection filters

- Corrected timestamp reconstruction in files containing more than one receiver time marker, in files recorded across a receiver restart, and in files where the SD card was pulled mid-deployment. Earlier versions could place reconstructed timestamps days to months away from the true time. If you produced corrected timestamps with v1.23–v1.26, reprocess those files.
- Fixed corrected dates printing with the year 0025.
- Added `Delta_Value` and `Internal_Value` diagnostic columns so you can see what the receiver logged next to what the filter reconstructed.
- Reorganized the Filter Options dialog into "Filter Methods" and "Data Quality / Verification", and relabeled the checkboxes. Behavior was unchanged by the renaming — see the Detection Filter History for the old and new names.

Receiver communication

- **Set Time** now also writes the receiver's timezone to match the time being set. Previously the clock and the timezone could disagree.
- Fixed GPS On/Off leaving the receiver half-on. The receiver has two separate GPS systems — the modem and sentence printing — and the buttons now track both.
- Fixed commands colliding in the receiver's buffer immediately after connecting, which could leave configuration fields blank.

- USB commands are now null-terminated, fixing occasional unreliable command delivery.

Other

- Added an **Auto** checkbox for USB connection, so automatic connection can be turned on and off instead of happening once per session. Automatic connection is suppressed during firmware flashing.
- The firmware download list now shows each file's release date.
- Firmware can be flashed without a connected receiver (settings validation is skipped and defaults are used).
- Fixed "Access is denied" errors writing to `Generated_Files` when the application was launched from a shortcut or an installed location.

v1.26 — May 2026

- Firmware download list reorganized into a grouped, sorted tree, with the most recent firmware in each folder marked. Legacy firmware folders start collapsed so they are only opened deliberately.
- Added a **Release Notes** button to the firmware download dialog, which opens the firmware version tracking document.
- Fixed firmware download failing on networks that inspect secure traffic (corporate proxy, VPN, or antivirus). Network errors now explain the actual cause — no connection, proxy, firewall, or GitHub rate limit — instead of always reporting "no internet".
- Added an **Auto** checkbox beside the Configuration tab's receiver time field that keeps the displayed receiver clock ticking once per second.
- Firmware flashing reliability: firmware generation is now detected and validated before any dialogs appear, so you are told what was found before choosing a file, rather than hitting an error after selecting one. Fixed several cases where flashing failed or crashed on both current and legacy firmware.

v1.25 — May 2026

- **Filtering extended to real-time logs** recorded by the interface, not just files downloaded from the SD card.
- Added the **Flag Out-of-Order Rows** filter.
- Added **Save Filtered Detections**: during a live session, a second CSV is written containing only detections matching your active filter.
- Downloaded SD card files are now saved exactly as the receiver wrote them, preserving the original headers and structure.
- Remote (TRC) downloads: added **Download all new files**, which queues every

completed file not yet saved locally and downloads them oldest-first; added **Sync Files** / **Stop Syncing**; added per-station **Hourly** and **Daily** schedule checkboxes and a configurable gap between downloads. Unattended downloads skip files already saved, and now correctly target the most recently *completed* file rather than the file still being written.

- Added the **Download Firmwares** button on the Configuration tab, which fetches firmware directly from the ATS public repository.
- Added a progress indicator during file transfers, a resizable About window with adjustable text size, and tooltips on every filter, graph, and file menu option.

v1.24 — March 2026

- Remote (TRC) command delivery reworked for reliability through the cellular modems used in field deployments.
- Cleaned up the remote file listing so SD card directory entries display correctly.

v1.23 — February 2026

- **Pressure sensor tag decoding finished** and usable on acoustically coded pressure tags.
- **Added time verification**, which cross-checks receiver timestamps against the receiver's internal counter and can optionally correct them.
- Time verification can be run on its own, without the code or false-positive filters.

v1.22 — November 2025

- Remote (TRC) file download completed for the most recent completed file.
- Added a diagnostic log file to help ATS troubleshoot reported issues.

v1.21 — November 2025

First production-ready release of the remote (TRC) tab, now visible to all users.

- Real-time detection logging from remote stations to a per-station CSV named by site and serial number.
- Fixed command reliability across different modem types.
- The detection table shows local time, sorts by detection count, and trims itself to stay responsive during long sessions.
- Log files roll over to a new file at 1 GB.

v1.18 — October 2025

- Firmware flashing cleaned up: clearer errors and warnings during a flash, and more reliable reconnection afterwards.

- The application now connects to an attached USB receiver on startup and pulls its configuration automatically.
- Added a visual indicator showing when the receiver is processing a command.
- Receiver settings and calibration offsets are restored after flashing newer firmware over legacy firmware.

v1.16 — October 2025

First release with post-processing filters.

- Added the **Code** filter and the **False Positive** (PRI) filter.
- Added the default `filter_output` folder.
- Added an interactive map of recorded GPS fixes.
- The USB driver installer is bundled with the application, and the correct driver is checked for on startup.
- Executables are digitally signed.

April–November 2025 — Initial development

First releases of the interface: USB and RS232 connection to a receiver, the live detection table with tag code summarizing, the command terminal, GPS fix parsing, SD card file download, firmware flashing, and the remote (TRC) tab for managing field-deployed receivers over cellular or Wi-Fi.

Questions

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